

Standard form for Logic Function

$$1/y = xy(z + \bar{z})$$

$$= \underline{xyz + xy\bar{z}}$$

$$f(xyz) = x\bar{y}z + x^0y + \underline{\bar{x}z}$$

$$\bar{x}z = \bar{x}z(y + \bar{y})$$

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Sum of Product

Standard form for Logic Function

1-SOP

minterm

$$f(x, y, z) = \underbrace{x \bar{y} z}_{\text{minterm}} + x y + \underline{\bar{x} z}$$

$$1y = xy(z + \bar{z})$$

$$= \underline{xyz} + x y \bar{z}$$

$$\bar{x} z = \bar{x} z (y + \bar{y})$$

$$F_{(xyz)} = \underset{\substack{1 \ 0 \ 1 \\ 5}}{X \bar{Y} Z} + \underset{\substack{1 \ 1 \ 1 \\ 7}}{X Y Z} + \underset{\substack{0 \ 1 \ 1 \\ 3}}{\bar{X} Y Z}$$

SOP

$$F_{(xyz)} = m_3 + m_5 + m_7$$

$$F_{(xyz)} = \underset{\substack{0 \ 0 \ 0 \\ 0}}{(X + Y + Z)} \underset{\substack{0 \ 0 \ 1 \\ 1}}{(X + Y + \bar{Z})} \underset{\substack{0 \ 1 \ 0 \\ 2}}{(X + \bar{Y} + Z)}$$

$$F_{(xyz)} = M_0 M_1 M_2$$

POS

22

sum-of-product

Standard form for Logic Function

 $m_0 + m_1 + m_2$

(4)

$$m_i = \overline{M}_i$$

1. SOP

$$f = \sum (m_i)$$

$$f = M_0 M_1 M_2$$

$$f = \prod (M_i)$$

$$\overline{xyz} = \overline{x} + \overline{y} + \overline{z}$$

$$\neq \overline{x + y + z}$$

$$= 100$$

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K-Map

$\bar{x}\bar{y}$

f_{xy}

$$f(xy) = \bar{x}\bar{y} + \bar{x}y + \underline{x\bar{y}} + \underline{xy}$$

f	$\bar{y}$		y	
	$\bar{x}$	x	$\bar{x}$	x
$\bar{x}$	1	0	1	0
x	0	1	1	1

$$f = \bar{x}$$

K-Map

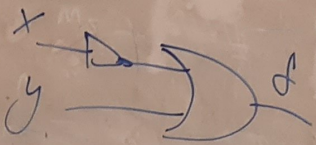
$\bar{x}\bar{y}$

f_{xy}

$f(xy) = \bar{x}\bar{y} + \bar{x}y + \underline{x\bar{y}} + \underline{xy}$

	$\bar{y}$	y
$\bar{x}$	1	1
x	0	1

$f = \bar{x} + y$



$f(x,y,z)$

3. variable

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{x}$	0	1	3	2
x	4	5	6	7

K-Map

$\bar{x}\bar{y}$

f_{xy}

	$\bar{a}\bar{b}$	$\bar{a}b$	$a\bar{b}$	ab
$\bar{a}$	1			
a				1

$f = \bar{c}$

$$f = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z +$$

$$\bar{x}y\bar{z} + \bar{x}yz$$

$$+ x\bar{y}\bar{z}$$

$f(x,y,z)$

$f = \bar{x} + \bar{y}\bar{z}$

3 variable

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{x}$	1	1	1	1
x	1			

Q:

Simplify the logic function and find the logic circuit:

$$f_{(xyz)} = M_0 + M_1 + M_2 M_3 + M_4 M_6$$